Ayush Jain

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PROFESSIONAL SUMMARY

Software engineer with experience shipping production backend systems and applied AI products through internships and projects. Hands-on experience across cloud infrastructure, APIs, data systems, and LLM applications, with a winning finish at an agentic AI hackathon involving 3,000+ builders.

EDUCATION

National University of Singapore — **Singapore**
Bachelor of Computing (Honours) in Computer Science — *Aug 2025 - Present*

- Completed NUS Overseas Colleges (NOC) Vietnam – Ho Chi Minh City.

Temasek Junior College — **Singapore**
Science Stream — *Jan 2021 - Dec 2022*

- Organised Building BloCS, an international outreach programme that introduced 100+ students to computing.

EXPERIENCE

Momos — **Ho Chi Minh City, Vietnam**
Software Engineering Intern (Backend) — *May 2026 - Jul 2026*

- Owned and shipped a backend project end-to-end in a production microservices codebase, from REST API and data-model design through implementation, testing and launch while aligning 5 cross functional teams on system ownership and rollout.
- Built and deployed an event-driven webhook integration across two TypeScript/Node.js services and a React frontend, configuring API Gateway throttling and JWT authentication before routing events through SQS to Lambda workers, automating ~10 hours of manual work across teams per week.
- Improved production reliability and performance through idempotent diff-based writes, scheduled reconciliation using cron jobs and DLQ monitoring with Datadog; optimised dependency loading in Lambda cold starts to maintain p99 API latency below 500 ms.
- Designed and iterated on the ERD and data model for a billing integration, modelling recovery paths, write ordering and soft-deleted records while selecting indexes based on expected query patterns.

Welyft Private Limited — **Singapore**
Software Engineer Intern — *Mar 2025 - Aug 2025*

- Designed and deployed an AWS EC2-hosted chatbot automation system, reducing manual scheduling effort by 60%+.
- Automated monthly CPF compliance workflows with Python, n8n, PGP encryption and SFTP, replacing repetitive file-preparation and secure-transfer steps.
- Built and shipped a responsive customer-facing landing site using TypeScript and Tailwind CSS.

PROJECTS & COMPETITIONS

BrainSquared | Agentic AI Build Week, Founders Track Winner — *Jul 2026*

- Won the Founders Mode Track as one of two winning teams among 3,000+ builders and 400+ projects.
- Co-built an organizational AI memory layer that turns fragmented prompts, workflows, agent setups and decisions into a reusable company knowledge graph.
- Designed an agentic pipeline for structured extraction, deduplication and standards validation, exposing governed organizational knowledge through MCP for reuse across AI workflows.

LearnGraph | OpenAI Build Week, Education Track — *Jul 2026*

- Led backend development of an adaptive learning platform that converts students' Obsidian notes into a concept graph and tracks understanding through evidence-based flashcards.
- Built the FastAPI/Postgres ingestion and retrieval pipeline, embedding concept nodes with pgvector for semantic similarity and combining embedding similarity with BM25 via reciprocal rank fusion for hybrid retrieval.
- Implemented RAG-based LLM grading with deterministic mastery updates and prerequisite-graph traversal, exposing live learner state through MCP for adaptive tutoring around demonstrated knowledge gaps.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL, C, HTML, CSS
Backend & Web: Node.js, Express.js, FastAPI, React, Next.js, REST APIs,
AI/ML: LLM applications, RAG, knowledge graphs, embeddings, hybrid retrieval, pgvector, BM25, reciprocal rank fusion,
Data & Cloud: PostgreSQL, MySQL, MongoDB, Supabase, AWS
DevOps & Testing: Terraform, Docker, CI/CD, Datadog, Git, GitHub, Jest & JUnit (unit, integration and end-to-end testing)